Macroeconomics #3

Ramsey–Cass–Koopmans Model

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Today's lecture

1. Introduction
2. The Ramsey model: setup, Euler equation, steady state
3. Dynamics and the phase diagram
4. Solving the model: guess and verify
5. Comparative dynamics
6. Ramsey vs. Solow

Introduction

From Solow to Ramsey

Recall the Solow model:

- The saving rate s is exogenous and constant.
- Households save a fixed fraction of income regardless of returns — why would agents not respond to incentives?

The Ramsey–Cass–Koopmans model (1928, 1965):

- A representative household optimally chooses consumption and saving over an infinite horizon.
- Saving is governed by an Euler equation $\Rightarrow$ micro-founded transition dynamics.
- Same type of steady state as Solow, but always dynamically efficient.

Roadmap (discrete time): Lagrangian $\Rightarrow$ Euler equation $\Rightarrow$ phase diagram and saddle path $\Rightarrow$ dynamic programming (guess and verify) for a closed-form solution.

Who was Frank Ramsey?



Frank P. Ramsey (1903–1930)

- British mathematician, philosopher and economist at Cambridge; Fellow of King's College at 21.
- Conversation with Keynes about economics, history, and mathematical philosophy.
- Wrote only three papers in economics, each founding a field:
  - *Truth and Probability* (1926): subjective probability.
  - *A Contribution to the Theory of Taxation* (1927): optimal taxation.
  - *A Mathematical Theory of Saving* (1928): optimal growth — today's model.
- Keynes: "one of the most remarkable contributions to mathematical economics ever made."
- Died in 1930, aged 26.

References

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The Ramsey Model

Household preferences

A representative household lives forever ($t = 0, 1, 2, \dots$) and values the consumption stream $\{c_t\}_{t=0}^{\infty}$ by

$$\sum_{t=0}^{\infty} \beta^t U(c_t)$$

where:

- c_t is per-capita consumption at time t.
- $\beta \in (0, 1)$ is the discount factor (patience parameter).
- $U(\cdot)$ is the instantaneous utility function.

Assumptions on U:

- $U'(c) > 0$ (more is better).
- $U''(c) < 0$ (diminishing marginal utility / desire to smooth consumption).
- Inada conditions: $\lim_{c \to 0} U'(c) = \infty$, $\lim_{c \to \infty} U'(c) = 0$.

CRRA utility

The most common specification is Constant Relative Risk Aversion (CRRA):

$$U(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \qquad \sigma > 0$$

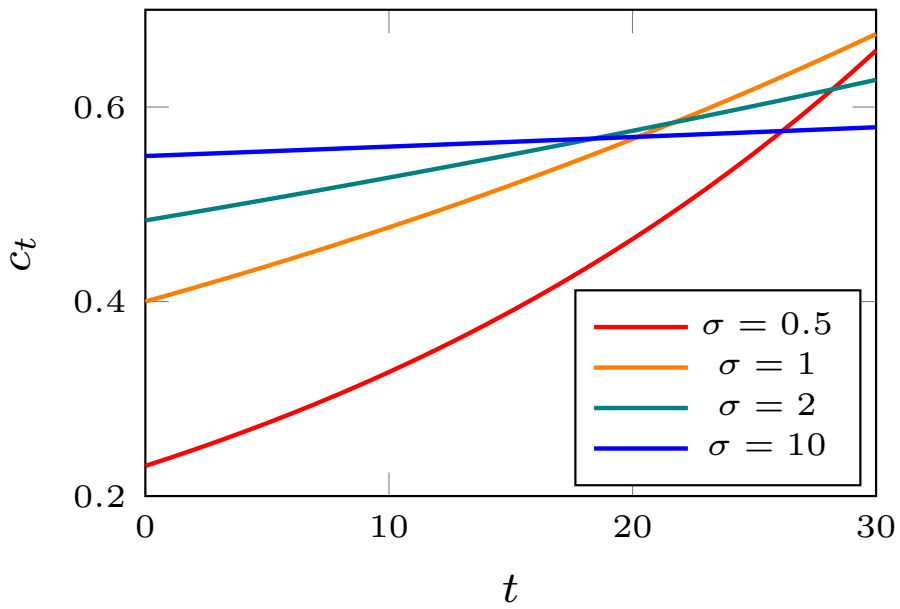
- σ is the coefficient of relative risk aversion: $-\frac{cU''(c)}{U'(c)} = \sigma$.
- $1/\sigma$ is the elasticity of intertemporal substitution (EIS).
- When $\sigma \to 1$: $U(c) = \ln(c)$.
- High σ: household dislikes consumption fluctuations (strong desire to smooth).
- Low σ: household is more willing to substitute consumption over time.

EIS and consumption smoothing

With CRRA utility and gross return R, the Euler equation gives

$$\frac{c_{t+1}}{c_t} = (\beta R)^{1/\sigma} \quad \Rightarrow \quad \text{EIS} \equiv \frac{d\ln(c_{t+1}/c_t)}{d\ln R} = \frac{1}{\sigma}$$

Same wealth, $\beta R = 1.018 > 1$



- Consumption smoothing: with $U'' < 0$, an even path beats an uneven one of equal present value.
- High σ (low EIS): path stays flat even when waiting pays.
- Low σ (high EIS): consume less today to exploit the return; path is steep.

Example: $\beta = 0.96$, $R = 1.06$, lifetime wealth $\sum_t c_t/R^t = 10$, so $c_0 = 10\,[1 - (\beta R)^{1/\sigma}/R]$.

Technology and resource constraint

Production:

$$Y_t = F(K_t, L_t), \qquad y_t = f(k_t) \equiv F(k_t, 1)$$

where f satisfies the neoclassical assumptions (CRS, $f' > 0$, $f'' < 0$, Inada).

In per-capita terms, the resource constraint is:

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t \quad \Longleftrightarrow \quad k_{t+1} - k_t = f(k_t) - \delta k_t - c_t$$

- Output is split between consumption and gross investment; $\delta \in [0, 1]$ is the depreciation rate.
- Net output $f(k_t) - \delta k_t$ that is not consumed adds to the capital stock.
- Solow: $c_t = (1-s)f(k_t)$ with s exogenous. Ramsey: c_t is chosen optimally.

The planner's problem

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(c_t)$$

subject to:

$$c_t + k_{t+1} = f(k_t) + (1-\delta)k_t, \qquad \forall\, t \geq 0$$

with $k_0 > 0$ given and the non-negativity constraints $c_t \geq 0$, $k_{t+1} \geq 0$.

Solution method: Form the Lagrangian.

Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ U(c_t) + \lambda_t \big[f(k_t) + (1-\delta)k_t - c_t - k_{t+1} \big] \Big\}$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 : \quad U'(c_t) = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 : \quad \lambda_t = \beta \lambda_{t+1} \big[f'(k_{t+1}) + 1 - \delta \big]$$

The multiplier $\lambda_t = U'(c_t)$ is the shadow price of resources at time t. The same FOC one period ahead gives

$$\lambda_{t+1} = U'(c_{t+1})$$

The Euler equation

Using the last three equations:

$$U'(c_t) = \beta\, U'(c_{t+1})\big[f'(k_{t+1}) + 1 - \delta\big]$$

Interpretation:

- LHS: Marginal utility cost of saving one more unit today.
- RHS: Discounted marginal utility gain from the extra return tomorrow.
- At the optimum, the household is indifferent between consuming today and saving for tomorrow.

Equivalently:

$$\frac{U'(c_t)}{\beta\, U'(c_{t+1})} = f'(k_{t+1}) + 1 - \delta \equiv R_{t+1}$$

The MRS equals the gross return on capital.

Euler equation: consumption dynamics

With CRRA utility $U(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, the Euler equation becomes:

$$c_t^{-\sigma} = \beta\, c_{t+1}^{-\sigma}\left[f'(k_{t+1}) + 1 - \delta\right]$$

Rearranging:

$$\frac{c_{t+1}}{c_t} = \left\{\beta\left[f'(k_{t+1}) + 1 - \delta\right]\right\}^{1/\sigma}$$

- Consumption grows when the net return exceeds the discount rate: $f'(k_{t+1}) + 1 - \delta > 1/\beta$.
- Consumption falls when returns are low: $f'(k_{t+1}) + 1 - \delta < 1/\beta$.
- Higher $\sigma \Rightarrow$ less responsive consumption growth (stronger smoothing motive).

Transversality condition (TVC)

The Euler equation alone admits multiple solutions. We need a boundary condition at infinity:

$$\lim_{t\to\infty} \beta^t U'(c_t)\, k_{t+1} = 0$$

Interpretation:

- The present value of capital left at the end of time must be zero.
- If $\lim \beta^t U'(c_t) k_{t+1} > 0$: the household is "dying rich" — it could increase utility by consuming more.
- The TVC rules out paths where capital grows too fast (inefficiently).

Together, the Euler equation + resource constraint + TVC fully characterize the optimal path $\{c_t, k_t\}_{t=0}^{\infty}$.

Optimal steady state

At the steady state, $c_t = c_{t+1} = c^*$ and $k_t = k_{t+1} = k^*$. The Euler equation gives:

$$1 = \beta\left[f'(k^*) + 1 - \delta\right] \quad \Rightarrow \quad f'(k^*) = \frac{1}{\beta} - 1 + \delta = \delta + \theta$$

where $\theta \equiv \frac{1-\beta}{\beta}$ is the rate of time preference. The resource constraint gives

$$c^* = f(k^*) - \delta k^*$$

Example ($f(k) = zk^\alpha$, any CRRA utility):

$$k^* = \left(\frac{\alpha z}{\delta + \theta}\right)^{\frac{1}{1-\alpha}}, \qquad c^* = z(k^*)^\alpha - \delta k^*$$

The steady state depends on β, δ and technology, but not on σ.

Comparative statics of the steady state

From $f'(k^*) = \delta + \theta$:

- More patient ($\beta \uparrow$, i.e. $\theta \downarrow$): $f'(k^*)$ falls $\Rightarrow k^* \uparrow \Rightarrow c^* \uparrow$.
  - Patient households save more, accumulate more capital.
- Higher depreciation ($\delta \uparrow$): $f'(k^*)$ rises $\Rightarrow k^* \downarrow \Rightarrow c^* \downarrow$.
- Higher TFP ($z \uparrow$ in $f(k) = zk^\alpha$): $k^* \uparrow \Rightarrow c^* \uparrow$.

Note: the risk aversion parameter σ does not affect the steady state — it only affects the speed of transition.

Modified Golden Rule and dynamic efficiency

- Every steady state satisfies $c = f(k) - \delta k$. The Golden Rule maximizes it:

$$f'(k^{gold}) = \delta$$

- The Ramsey steady state satisfies the Modified Golden Rule:

$$f'(k^*) = \delta + \theta > \delta = f'(k^{gold}) \quad \Rightarrow \quad k^* < k^{gold}$$

- Why stop short of k^{gold}? Getting there requires giving up consumption today; an impatient household ($\beta < 1$) values the transition path, not just the steady state.
- In Solow, a "too high" s can push $k > k^{gold}$ — dynamic inefficiency. In Ramsey this never happens: the TVC rules out over-accumulation.

Dynamics and Phase Diagram

The dynamic system

The optimal path $\{c_t, k_t\}$ solves two difference equations:

$$U'(c_t) = \beta\, U'(c_{t+1})\big[f'(k_{t+1}) + 1 - \delta\big] \qquad \text{(Euler)}$$

$$k_{t+1} = f(k_t) + (1-\delta)k_t - c_t \qquad \text{(Resource)}$$

with k_0 given and the TVC. For example, with log utility the Euler equation is

$$c_{t+1} = \beta c_t\big[f'(k_{t+1}) + 1 - \delta\big]$$

To draw the phase diagram, write the system in changes

$$\Delta c_t = c_{t+1} - c_t, \qquad \Delta k_t = k_{t+1} - k_t$$

and find the loci where each change is zero.

Phase diagram: the $\Delta k = 0$ locus

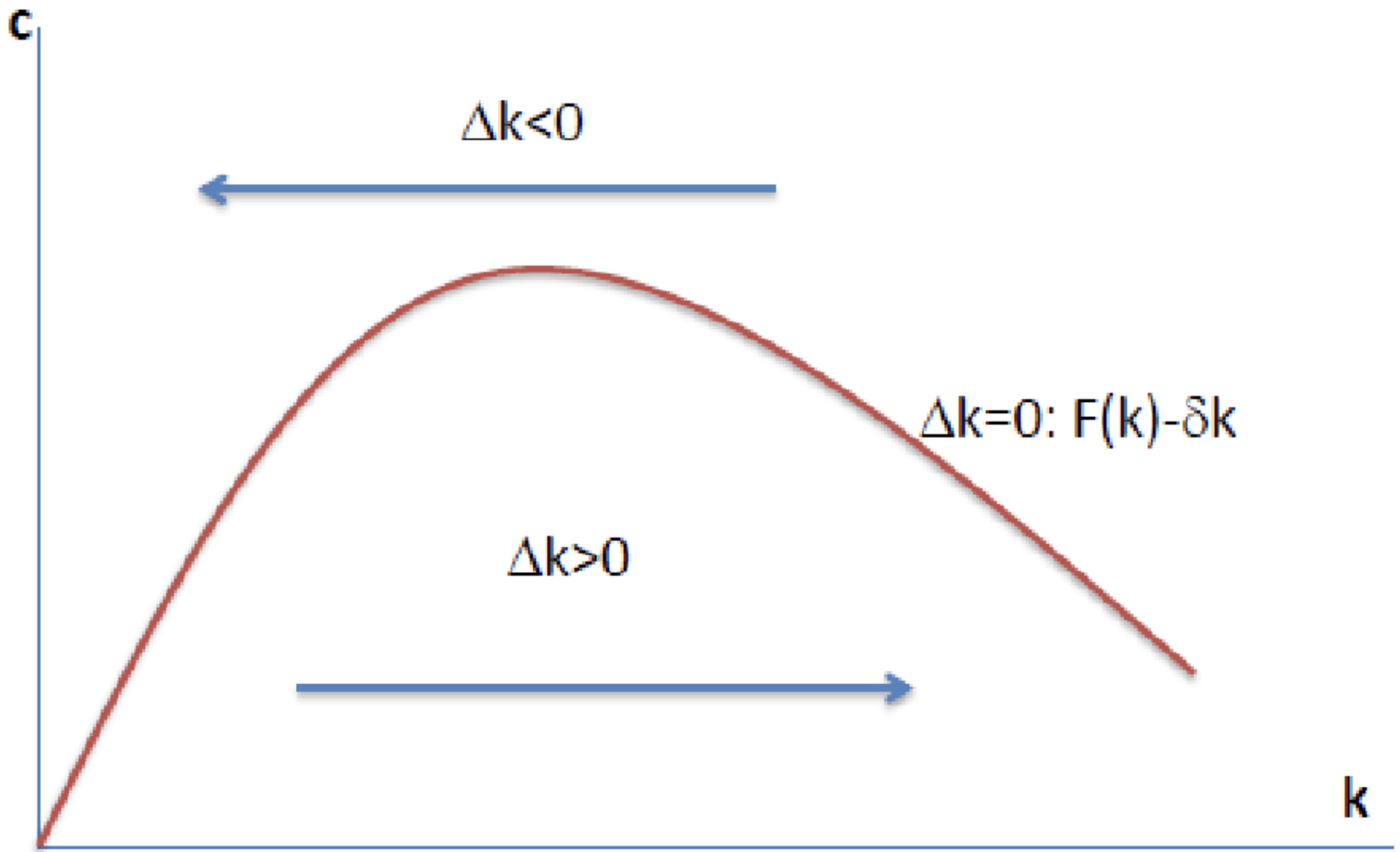
From the resource constraint:

$$\Delta k \equiv k_{t+1} - k_t = f(k_t) - \delta k_t - c_t$$

The $\Delta k = 0$ locus is:

$$c = f(k) - \delta k$$

- This is a hump-shaped curve (why?).
- Above the locus: $c > f(k) - \delta k \Rightarrow \Delta k < 0$ (capital falls).
- Below the locus: $c < f(k) - \delta k \Rightarrow \Delta k > 0$ (capital rises).



Dynamics of capital

Phase diagram: the $\Delta c = 0$ locus

From the Euler equation, $c_{t+1} = c_t$ when:

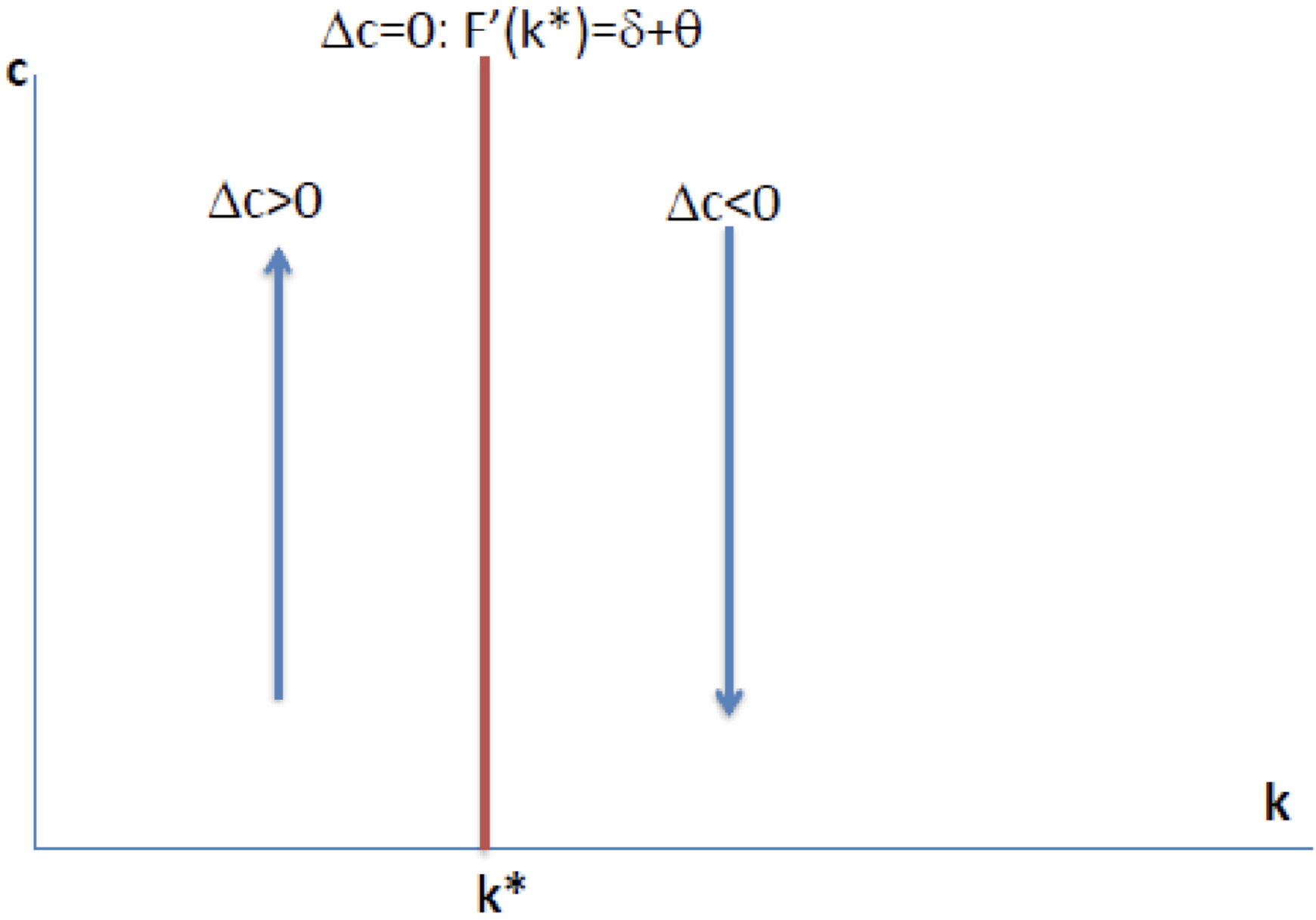
$$\beta\big[f'(k) + 1 - \delta\big] = 1 \quad \Rightarrow \quad f'(k^*) = \delta + \theta$$

So $c_{t+1} = c_t$ iff $k_{t+1} = k^*$. Using $k_{t+1} = f(k_t) + (1-\delta)k_t - c_t$, the exact $\Delta c = 0$ locus is

$$c = f(k) + (1-\delta)k - k^*,$$

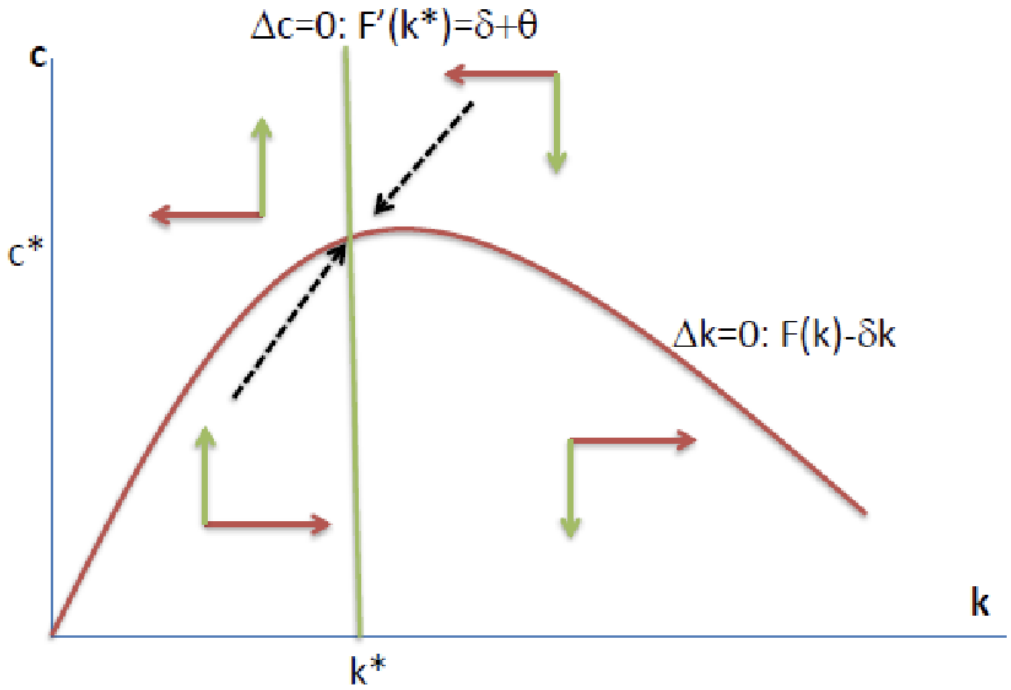
a steep upward-sloping curve through (k^*, c^*). In continuous time (or with short periods) it becomes the vertical line $k = k^*$, which is how it is drawn in the figures.
Pay attention to the sign of Δc when k changes

- If $k_{t+1} < k^*$ (i.e. c_t above the locus): $f'(k_{t+1}) > \delta + \theta \Rightarrow$ returns are high $\Rightarrow \Delta c > 0$ (consumption rising).
- If $k_{t+1} > k^*$ (i.e. c_t below the locus): $f'(k_{t+1}) < \delta + \theta \Rightarrow$ returns are low $\Rightarrow \Delta c < 0$ (consumption falling).



Dynamics of consumption

Putting together: the phase diagram



Phase diagram: dynamics of consumption and capital

- The two loci divide the (k, c) plane into four regions.
- In each region, arrows indicate the direction of motion.
- The steady state (k^*, c^*) is at the intersection.

Saddle-path stability

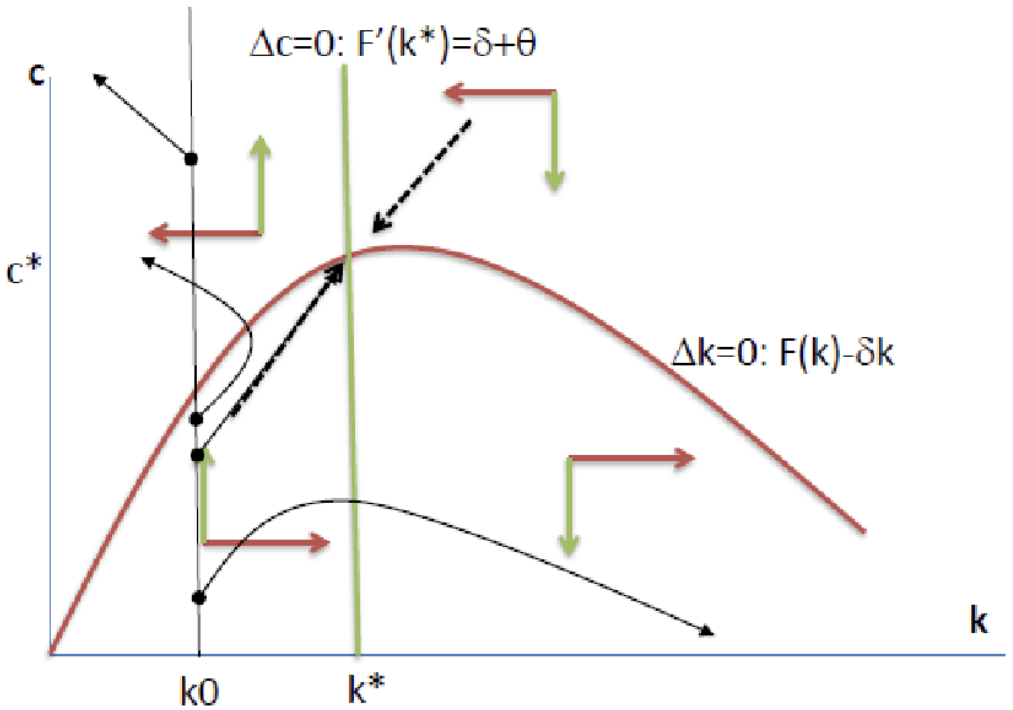
- Given any initial (k_0, c_0), the phase diagram traces the path $\{k_t, c_t\}$.
- k_0 is predetermined (state variable); c_0 is free and can jump (control variable).
- For a given k_0, there is a unique c_0 such that the economy converges to (k^*, c^*). This trajectory is the saddle path.
- The system has one stable and one unstable eigenvalue, so the saddle path (stable arm) is one-dimensional.

Any other c_0 fails:

- Above the saddle path: capital is run down to zero and consumption collapses (infeasible).
- Below the saddle path: capital is over-accumulated, violating the TVC.

Hence optimal consumption is a function of capital, $c_t = h(k_t)$: the saddle path is the policy function.

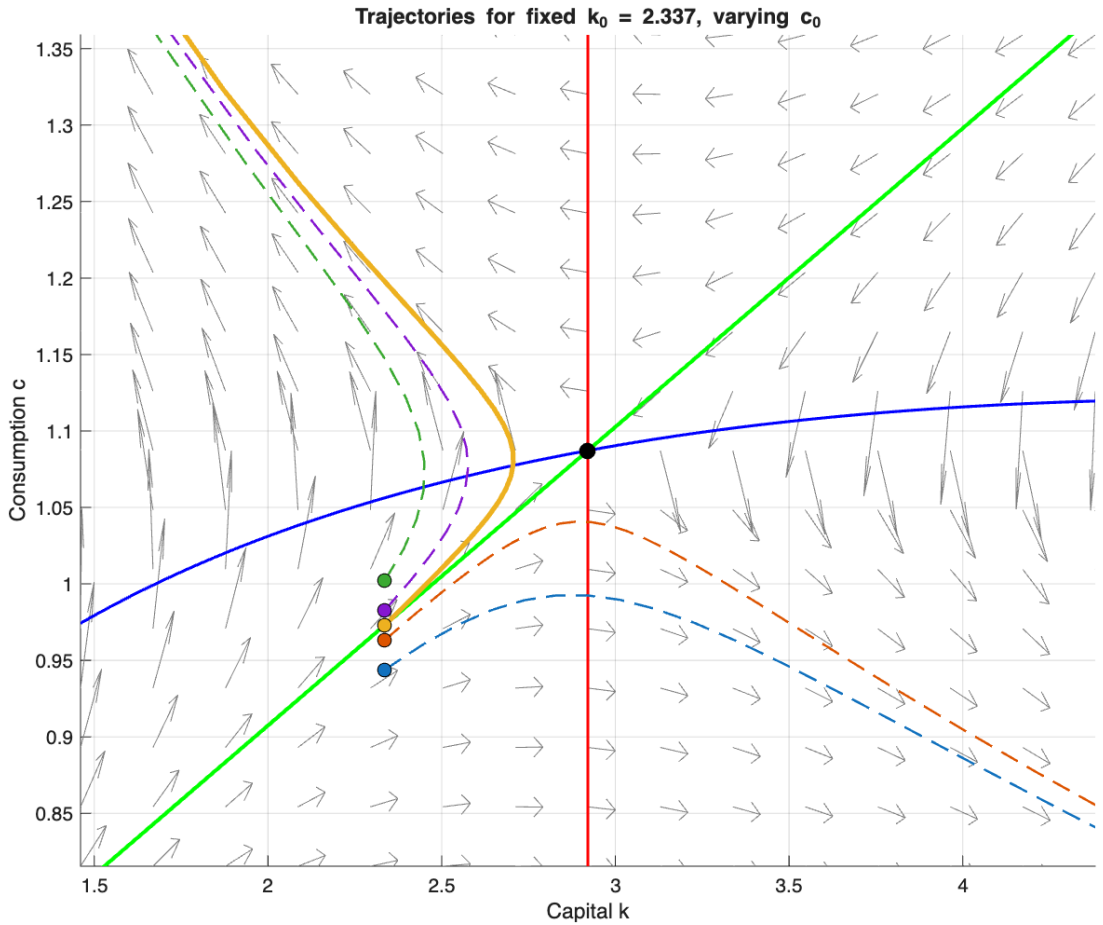
The saddle path in the phase diagram



Phase diagram: saddle path and dynamics from different initial consumption levels

- At $t = 0$, c jumps onto the saddle path above k_0.
- The economy then follows the saddle path to the steady state.

Numerical phase diagram



Trajectories starting exactly on the green line (the linearised saddle path) converge to the steady state. Those starting slightly off it diverge, because the unstable component grows exponentially.

Solving the Ramsey Model: Guess and Verify

Recursive formulation of the Ramsey problem

As in the cake-eating problem, define the value function $V(k)$: the maximum lifetime utility attainable from capital stock k.

The Bellman equation:

$$V(k) = \max_{0 \le k' \le f(k)+(1-\delta)k} \left\{ U\big(f(k) + (1-\delta)k - k'\big) + \beta V(k') \right\}$$

- State variable: k (capital today).
- Choice variable: k' (capital tomorrow); consumption is $c = f(k) + (1-\delta)k - k'$.
- Cake-eating is the special case $f(k) = 0$, $\delta = 0$: the cake does not grow.

The solution is a policy function $k' = g(k)$ (equivalently $c = h(k)$, the saddle path), together with $V(k)$.

FOC, envelope condition and the Euler equation

FOC with respect to k':

$$U'(c) = \beta V'(k')$$

Envelope condition (differentiate V with respect to the state k):

$$V'(k) = U'(c)\left[f'(k) + 1 - \delta\right]$$

Shift the envelope condition one period forward:

$$V'(k') = U'(c')\left[f'(k') + 1 - \delta\right]$$

Combine with the FOC:

$$U'(c) = \beta\, U'(c')\left[f'(k') + 1 - \delta\right]$$

The same Euler equation as the Lagrangian approach. But now we have a recipe to find the policy function itself.

A case with a closed-form solution

Guess and verify works only for special cases. The best-known one is Brock and Mirman (1972):

$$U(c) = \ln c, \qquad f(k) = zk^{\alpha}, \qquad \delta = 1$$

The resource constraint becomes

$$c_t + k_{t+1} = zk_t^{\alpha}$$

The Bellman equation:

$$V(k) = \max_{k'} \left\{ \ln \left(zk^{\alpha} - k' \right) + \beta V(k') \right\}$$

- Why $\delta = 1$? With $\delta < 1$, the term $(1 - \delta)k$ breaks the log-linear structure and no closed form exists.
- Full depreciation is a reasonable approximation when a period is long (e.g. a generation).

Guess and verify the value function (1/2)

Guess: as in cake-eating, the value function is log-linear:

$$V(k) = A + B \ln k$$

The Bellman equation becomes

$$A + B \ln k = \max_{k'} \left\{ \ln \left(z k^{\alpha} - k' \right) + \beta \left[A + B \ln k' \right] \right\}$$

FOC with respect to k':

$$\frac{1}{z k^{\alpha} - k'} = \frac{\beta B}{k'}$$

implying

$$k' = \frac{\beta B}{1 + \beta B} z k^{\alpha}, \qquad c = z k^{\alpha} - k' = \frac{1}{1 + \beta B} z k^{\alpha}$$

Guess and verify the value function (2/2)

Substitute c and k' back into the Bellman equation:

$$\begin{aligned} A + B\ln k = {} & \ln z + \alpha \ln k - \ln(1+\beta B) \\ & + \beta A + \beta B\Big[\ln \tfrac{\beta B}{1+\beta B} + \ln z + \alpha \ln k\Big] \end{aligned}$$

Matching coefficients:

$$(\ln k): \quad B = \alpha + \alpha\beta B \quad \Longrightarrow \quad B = \frac{\alpha}{1-\alpha\beta}$$

$$(\text{const.}): \quad A = \frac{1}{1-\beta}\left[\ln(1-\alpha\beta) + \frac{\alpha\beta}{1-\alpha\beta}\ln(\alpha\beta) + \frac{\ln z}{1-\alpha\beta}\right]$$

Since $\beta B/(1+\beta B) = \alpha\beta$, the policy functions are

$$\boxed{k' = \alpha\beta\, z k^{\alpha}, \qquad c = (1-\alpha\beta)\, z k^{\alpha}}$$

The guess is verified: both sides of the Bellman equation are of the form $A + B\ln k$.

Guess and verify with the Euler equation

Alternatively, skip the value function and guess the policy function directly:

$$k_{t+1} = s\, z k_t^{\alpha}, \qquad c_t = (1-s)\, z k_t^{\alpha}$$

The Euler equation with $\delta = 1$:

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} \alpha z k_{t+1}^{\alpha-1}$$

Plug in the guess (use $c_{t+1} = (1-s) z k_{t+1}^{\alpha}$):

$$\text{RHS} = \frac{\alpha\beta\, z k_{t+1}^{\alpha-1}}{(1-s)\, z k_{t+1}^{\alpha}} = \frac{\alpha\beta}{(1-s)\, k_{t+1}} = \frac{\alpha\beta}{(1-s)\, s\, z k_t^{\alpha}}$$

$$\text{LHS} = \frac{1}{(1-s)\, z k_t^{\alpha}}$$

LHS = RHS for all k_t if and only if $s = \alpha\beta$. Same answer as before.

Checking the transversality condition

The Euler equation alone does not pin down the solution (recall the half-cake plan). We must check the TVC:

$$\lim_{t\to\infty} \beta^t U'(c_t)\, k_{t+1} = \lim_{t\to\infty} \beta^t \frac{\alpha\beta\, y_t}{(1-\alpha\beta)\, y_t} = \frac{\alpha\beta}{1-\alpha\beta} \lim_{t\to\infty} \beta^t = 0 \quad \checkmark$$

- Any other constant saving rate $s \neq \alpha\beta$ violates the Euler equation.
- Paths that satisfy the Euler equation but start from a different c_0 violate either feasibility ($k \to 0$) or the TVC.
- Hence the policy $c = (1-\alpha\beta)zk^\alpha$ is the saddle path of the phase diagram, in closed form.

Interpretation: Ramsey meets Solow

$$k_{t+1} = \alpha\beta\, y_t, \qquad c_t = (1 - \alpha\beta)\, y_t$$

- The household saves a constant fraction $s^* = \alpha\beta$ of output.
- This is exactly the Solow model (with $\delta = 1$) with saving rate $s = \alpha\beta$ — but now s is derived, not assumed.
- More patient ($\beta \uparrow$) $\Rightarrow$ higher saving rate.
- Higher capital share ($\alpha \uparrow$) $\Rightarrow$ capital is more productive $\Rightarrow$ higher saving rate.
- TFP z does not affect the saving rate: with log utility, income and substitution effects of a higher return cancel out.

In general (CRRA with $\sigma \neq 1$, or $\delta < 1$), the optimal saving rate varies along the transition — and the saddle path must be found numerically.

Steady state and transition dynamics

The law of motion $k_{t+1} = \alpha\beta z k_t^{\alpha}$ is linear in logs:

$$\ln k_{t+1} = \ln(\alpha\beta z) + \alpha \ln k_t$$

Steady state:

$$k^* = (\alpha\beta z)^{\frac{1}{1-\alpha}}, \qquad c^* = (1 - \alpha\beta)\, z (k^*)^{\alpha}$$

Check: $f'(k^*) = \alpha z (k^*)^{\alpha-1} = 1/\beta = \delta + \theta$ with $\delta = 1$. ✓

Solving the difference equation:

$$\ln k_t - \ln k^* = \alpha^t \big(\ln k_0 - \ln k^* \big)$$

- Since $0 < \alpha < 1$: monotone convergence to k^* from any $k_0 > 0$.
- The gap (in logs) shrinks by the factor α each period; half-life $= \ln 0.5 / \ln \alpha$.

Numerical example

Parameters: $\alpha = 0.3$, $\beta = 0.95$, $z = 1$, starting from $k_0 = 0.1\,k^*$.

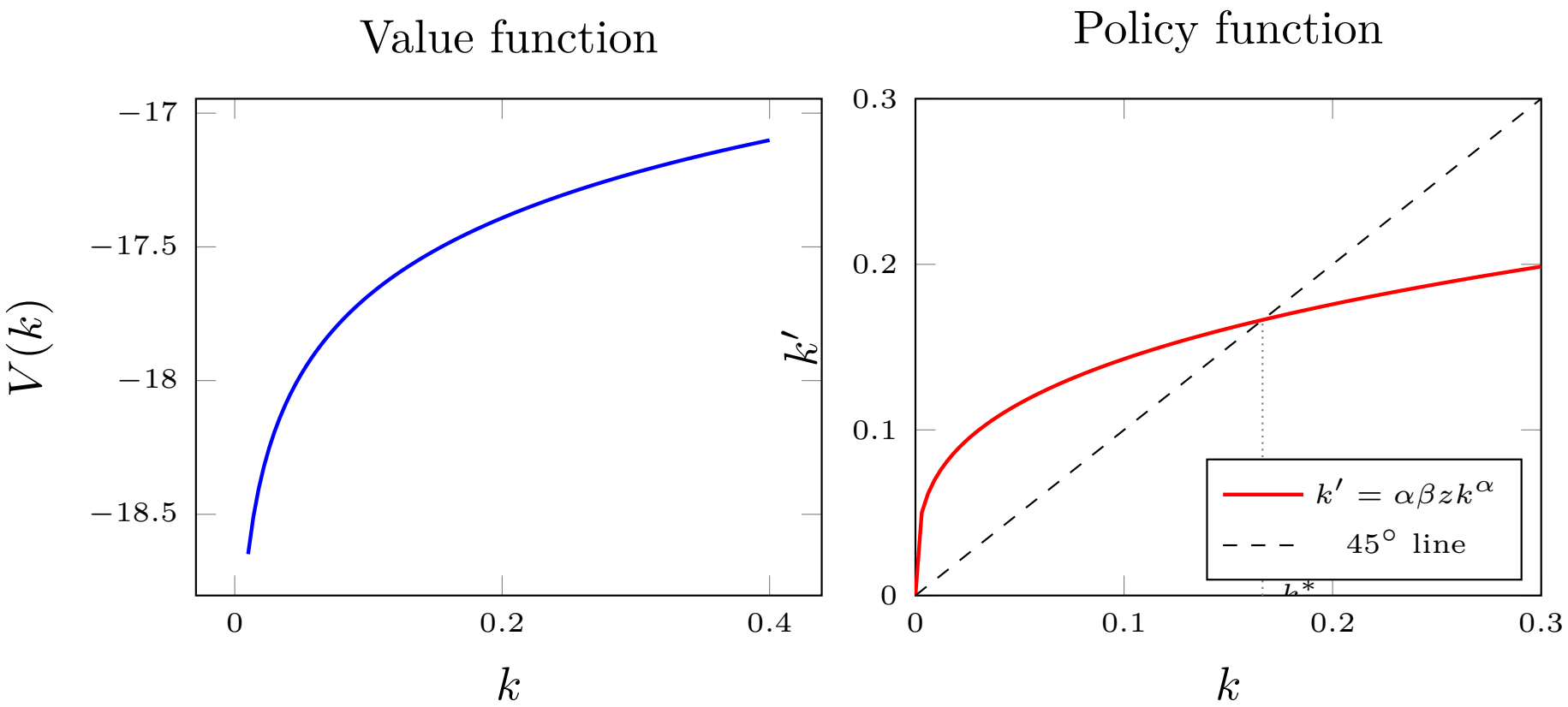
$$s^* = \alpha\beta = 0.285, \quad k^* = 0.1664, \quad y^* = 0.5839, \quad c^* = 0.4175$$

Iterate $y_t = zk_t^\alpha$, $c_t = (1-\alpha\beta)y_t$, $k_{t+1} = \alpha\beta\, y_t$:

t	k_t	y_t	c_t	k_{t+1}
0	0.0166	0.2927	0.2093	0.0834
1	0.0834	0.4746	0.3394	0.1353
2	0.1353	0.5487	0.3923	0.1564
3	0.1564	0.5731	0.4098	0.1633
4	0.1633	0.5807	0.4152	0.1655
5	0.1655	0.5830	0.4168	0.1661
⋮	⋮	⋮	⋮	⋮
∞	0.1664	0.5839	0.4175	0.1664

One can check that the Euler equation holds exactly between every pair of consecutive periods.

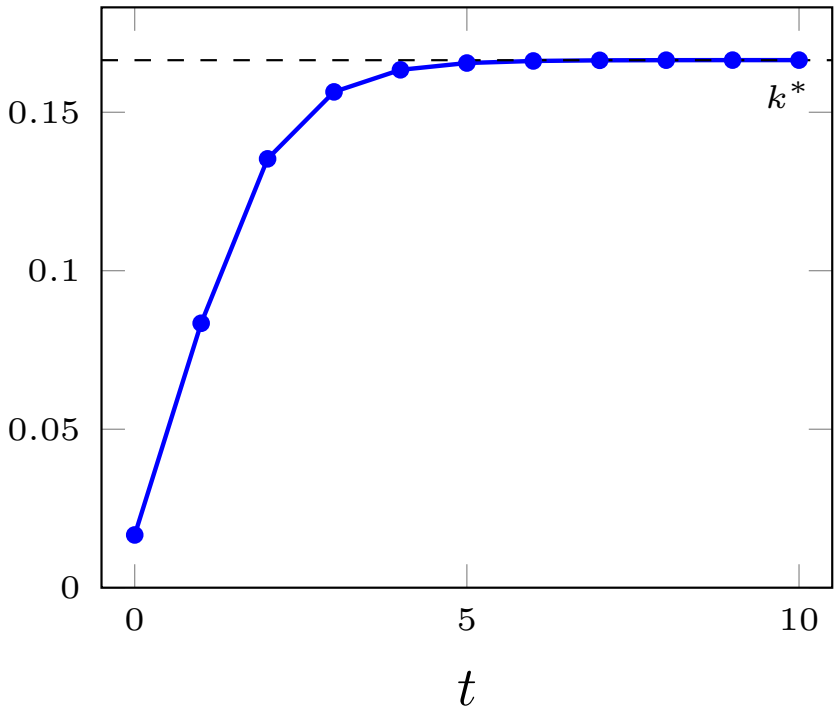
Value and policy functions



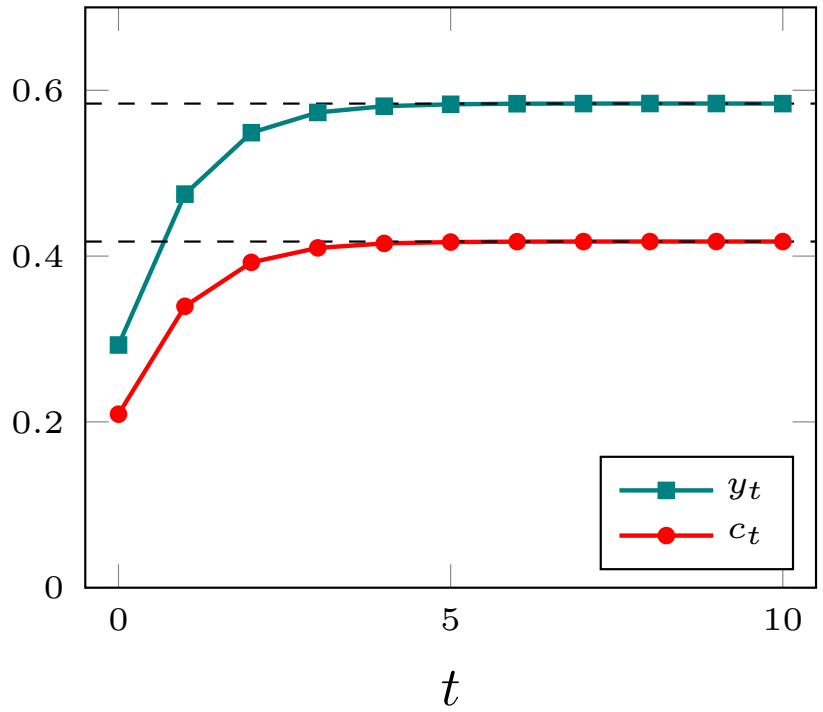
- $V(k) = A + B \ln k$ is increasing and concave, as in cake-eating.
- The policy function crosses the 45° line once at k^* from above: $k < k^* \Rightarrow k' > k$.

Simulation: from k_0 to the steady state

Capital k_t

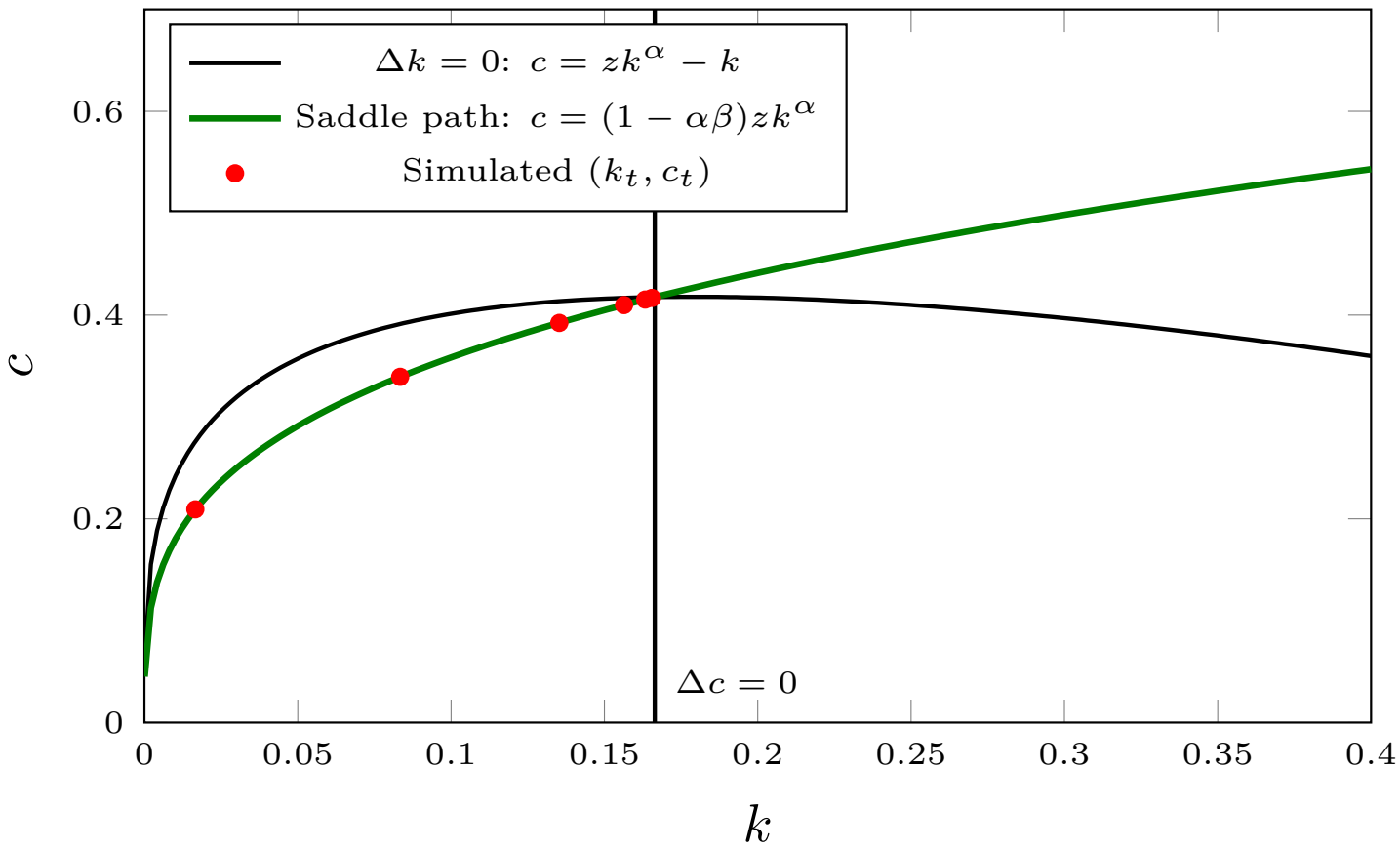


Output y_t and consumption c_t



- Starting below k^*, capital, output and consumption all rise monotonically.
- Convergence is fast here because $\delta = 1$ (a period is long) and $\alpha = 0.3$.

The simulated path in the phase diagram



- The optimal policy function is the saddle path: given k_0, c_0 jumps onto it and the economy slides to (k^*, c^*).

Beyond the closed form

With $\delta < 1$ or CRRA utility ($\sigma \neq 1$), there is no closed-form $V(k)$. We solve numerically by value function iteration:

1. Build a grid for k around k^* and start from any guess $V_0(k)$ (e.g. $V_0 = 0$).
2. Update: $V_{n+1}(k) = \max_{k'} \{U(f(k) + (1-\delta)k - k') + \beta V_n(k')\}$.
3. Repeat until $\max_k |V_{n+1}(k) - V_n(k)|$ is small.
4. Read off the policy $k' = g(k)$, then simulate $k_{t+1} = g(k_t)$ from k_0.

- Convergence is guaranteed: the Bellman operator is a contraction (Blackwell's sufficient conditions).
- Guess and verify is the special case where we can solve the fixed point $V = TV$ with pen and paper.

Comparative Dynamics

Comparative dynamics: increase in θ (less patience)

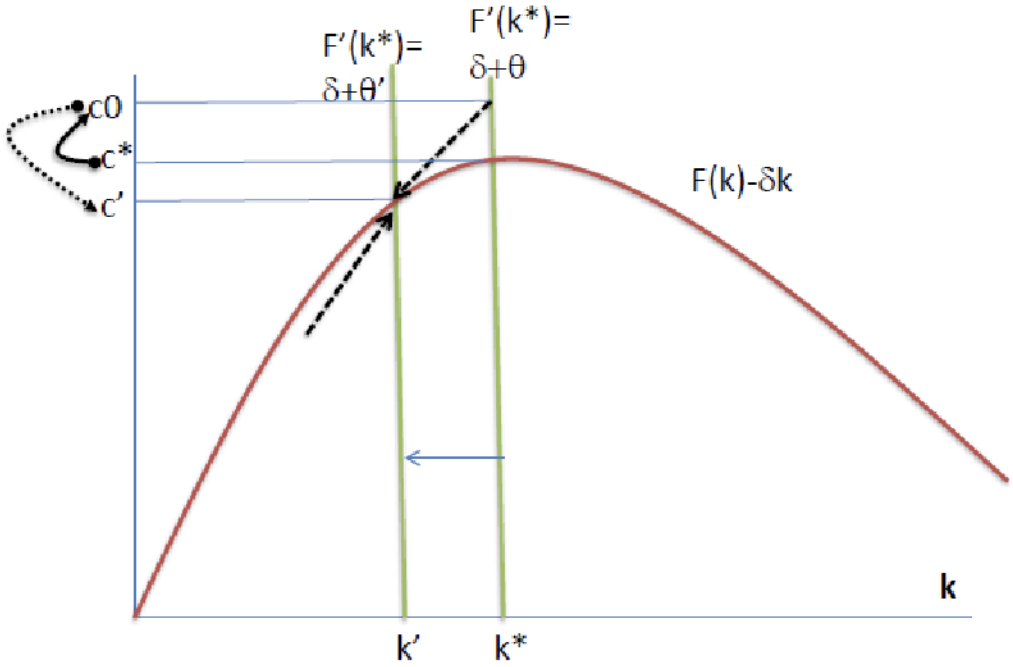
Suppose households become less patient: θ increases (β falls).

- The $\Delta c = 0$ locus: $f'(k^*) = \delta + \theta$ shifts to the left (lower k^*; in discrete time the curve $c = f(k) + (1-\delta)k - k^*$ shifts up).
- The $\Delta k = 0$ locus: $c = f(k) - \delta k$ is unaffected.

Dynamics:

1. When θ increases, c_t jumps up to the new saddle path.
2. In the short run: higher consumption, lower investment.
3. In the long run: capital falls to the new (lower) k^*, and consumption falls to the new (lower) c^*.

Phase diagram: increase in θ



Phase diagram: An increase in the discount rate (θ).

- Short run: c_0 jumps up $\Rightarrow$ saving falls.
- Transition: k and c both decline along the new saddle path.
- Long run: lower k^*, lower c^*.

Comparative dynamics: improvement in technology

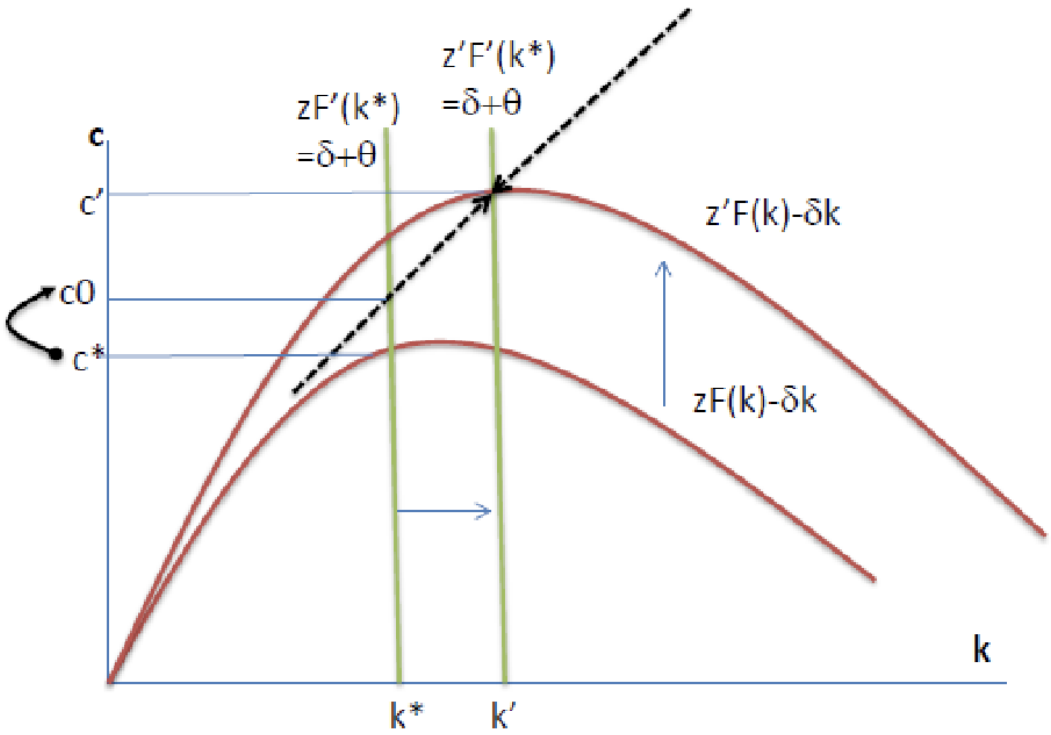
Suppose TFP increases: $z \uparrow$ (so $f(k) = zk^{\alpha}$ shifts up).

- $\Delta c = 0$: $\alpha z (k^*)^{\alpha - 1} = \delta + \theta \Rightarrow$ shifts to the right (higher k^*).
- $\Delta k = 0$: $c = zk^{\alpha} - \delta k$ shifts up (higher net output).
- In the long run: both k^* and c^* increase.

But what happens in the short run? It depends on σ:

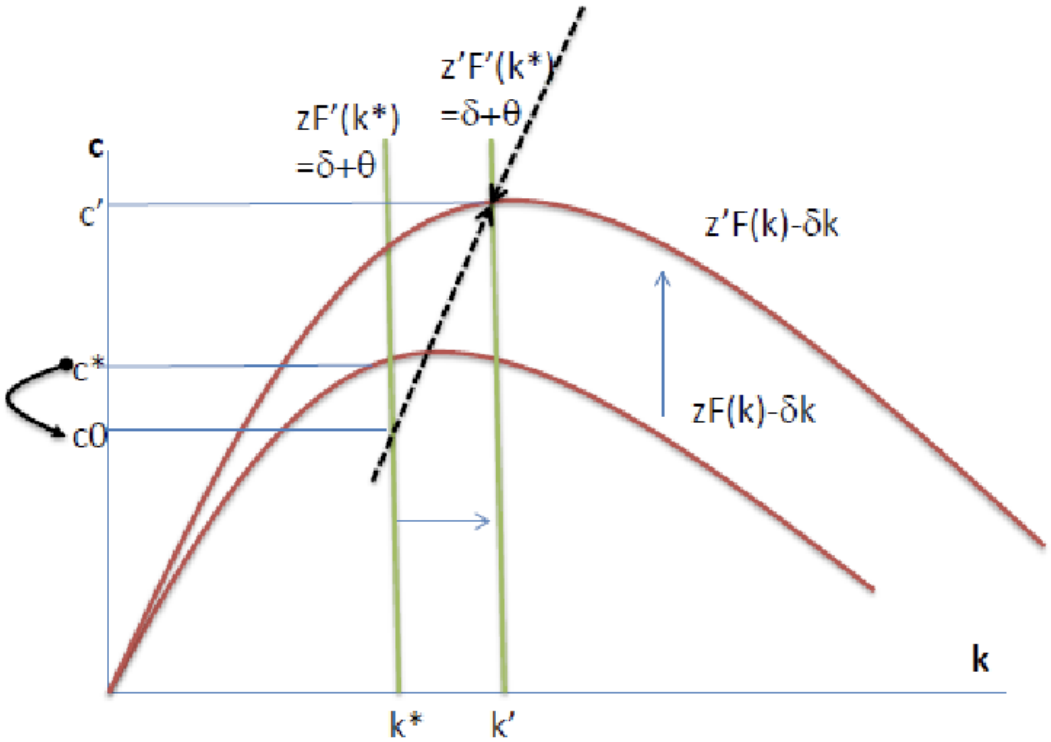
- High σ (low EIS, strong smoothing): c_0 jumps up; slow transition.
- Low σ (high EIS, willing to substitute): c_0 may fall; fast capital accumulation.

Technology shock: low EIS (high σ)



- High σ (low EIS): the household strongly dislikes uneven consumption paths; the wealth effect of the shock dominates the substitution effect.
- Household wants to smooth consumption $\Rightarrow$ c_0 increases immediately.
- Capital accumulates slowly to the new steady state.

Technology shock: high EIS (low σ)



- Low σ (high EIS): the household is willing to tilt consumption towards the future; the substitution effect of the higher return dominates.
- Households want to forgo consumption, invest more to reach the new steady state more quickly.
- c_0 may actually decrease in the short run.

Ramsey vs. Solow

What does Ramsey add to Solow?

	Solow	**Ramsey**
Saving rate	Exogenous s	Endogenous
Micro-foundations	No	Yes
Dynamic efficiency	Not guaranteed	Always
Golden Rule	Possible to overshoot	No overshooting
Transition dynamics	Monotone convergence	Saddle-path convergence

- Ramsey's main contribution: endogenous saving and dynamic efficiency.
- Both models share the same limitation: no endogenous long-run growth.

Exercises

Exercise 1: Euler equation and steady state

Consider the planner's problem with $U(c) = \ln(c)$ and $f(k) = k^{\alpha}$.

1. Write down the Lagrangian.
2. Derive the Euler equation.
3. Find the steady-state k^* and c^* as functions of α, β, and δ.
4. How does k^* change when β increases? Interpret.

▸ Solution

Exercise 2: Phase diagram analysis

Consider the model with $U(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, $f(k) = k^{\alpha}$, $\alpha = 0.3$, $\delta = 0.05$, $\beta = 0.96$, $\sigma = 2$.

1. Compute k^*, c^*, and k^{gold}.
2. Sketch the phase diagram with the $\Delta c = 0$ and $\Delta k = 0$ loci.
3. Starting from $k_0 = k^*/2$, describe the transition dynamics qualitatively.
4. What happens if β decreases permanently? Sketch the new phase diagram.

▸ Solution

Solution to Exercise 1

With $U(c) = \ln c$ and $f(k) = k^\alpha$:

1. $\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \{\ln c_t + \lambda_t [k_t^\alpha + (1-\delta)k_t - c_t - k_{t+1}]\}$.
2. FOCs: $1/c_t = \lambda_t$ and $\lambda_t = \beta \lambda_{t+1} [\alpha k_{t+1}^{\alpha-1} + 1 - \delta]$.

   Euler equation: $\frac{1}{c_t} = \frac{\beta}{c_{t+1}} [\alpha k_{t+1}^{\alpha-1} + 1 - \delta]$.
3. At steady state: $\alpha (k^*)^{\alpha-1} = 1/\beta - 1 + \delta$, so

$$k^* = \left(\frac{\alpha}{1/\beta - 1 + \delta} \right)^{1/(1-\alpha)}, \quad c^* = (k^*)^\alpha - \delta k^*.$$

4. $\beta \uparrow \Rightarrow$ denominator $\downarrow \Rightarrow k^* \uparrow$. More patient households save more.

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Solution to Exercise 2

With $\alpha = 0.3$, $\delta = 0.05$, $\beta = 0.96$, so $\theta = 1/\beta - 1 \approx 0.0417$:

1. $f'(k^*) = \delta + \theta$: $k^* = \left(\frac{0.3}{0.0917}\right)^{10/7} \approx 5.44$.
   $c^* = (k^*)^{0.3} - 0.05 \times k^* \approx 1.66 - 0.27 = 1.39$.
   $f'(k^{gold}) = \delta$: $k^{gold} = (0.3/0.05)^{10/7} = 6^{10/7} \approx 12.93 > k^*$. ✓
2. Vertical $\Delta c = 0$ at k^*; hump-shaped $\Delta k = 0$ curve $c = k^\alpha - \delta k$; saddle path through (k^*, c^*).
3. From $k_0 < k^*$: c_0 jumps onto the saddle path, then both c and k increase monotonically to (k^*, c^*).
4. $\beta \downarrow$ ($\theta \uparrow$): $\Delta c = 0$ shifts left, $\Delta k = 0$ unchanged. c jumps up initially, then both c and k decline to the new lower steady state.

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